Liceria & Co.

All About Pizza!

SQL PROJECT ON PIZZA SALES

Hello, my name is Rahul Dolai. In this project, I have utilized SQL queries to solve business questions related to pizza sales. This project demonstrates my SQL skills in data analysis, including data exploration, sales analysis, and generating business insights.

ALL QUESTIONS

Basic:--

1.Retrieve the total number of orders placed.
2.Calculate the total revenue generated from pizza sales.
3.Identify the highest-priced pizza.
4.Identify the most common pizza size ordered.
5.List the top 5 most ordered pizza types along with their quantities.

Intermediate:--

6.Join the necessary tables to find the total quantity of each pizza category.
7.Determine the distribution of orders by hour of the day.
8.Join relevant tables to find the category-wise distribution of pizzas.
9.Group the orders by date and calculate the average number of pizzas ordered per day.
10.Determine the top 3 most ordered pizza types based on revenue.

Advanced:--

11.Calculate the percentage contribution of each pizza type to total revenue.
12.Analyze the cumulative revenue generated over time.
13.Determine the top 3 most ordered pizza types based on revenue for each pizza category.

1.RETRIEVE THE TOTAL NUMBER OF ORDERS PLACED.

```sql
SELECT
    COUNT(order_id) AS Total_Orders
FROM
    orders;
```

	Total_Orders
▶	21350

2.CALCULATE THE TOTAL REVENUE GENERATED FROM PIZZA SALES.

```
SELECT
    ROUND(SUM(p.price * o.quantity), 2) AS Total_Revenue
FROM
    pizzas p
        JOIN
    order_details o ON p.pizza_id = o.pizza_id;
```

Total_Revenue
817860.05

3.IDENTIFY THE HIGHEST-PRICED PIZZA.

```sql
SELECT
    pt.name, p.price
FROM
    pizzas p
        JOIN
    pizza_types pt ON pt.pizza_type_id = p.pizza_type_id
ORDER BY p.price DESC
LIMIT 1;
```

name	price
The Greek Pizza	35.95

4.IDENTIFY THE MOST COMMON PIZZA SIZE ORDERED.

```sql
SELECT
    p.size, COUNT(o.order_details_id) AS Order_Count
FROM
    pizzas p
        JOIN
    order_details o ON o.pizza_id = p.pizza_id
GROUP BY p.size
ORDER BY Order_Count DESC;
```

size	Order_Count
L	18526
M	15385
S	14137
XL	544
XXL	28

5.LIST THE TOP 5 MOST ORDERED PIZZA TYPES ALONG WITH THEIR QUANTITIES.

```
SELECT
    pt.name, SUM(o.quantity) AS Total_Quantity
FROM
    pizza_types pt
        JOIN
    pizzas p ON pt.pizza_type_id = p.pizza_type_id
        JOIN
    order_details o ON p.pizza_id = o.pizza_id
GROUP BY pt.name
ORDER BY Total_Quantity DESC
LIMIT 5;
```

name	Total_Quantity
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

6.JOIN THE NECESSARY TABLES TO FIND THE TOTAL QUANTITY OF EACH PIZZA ORDERED.

```sql
SELECT
    p.pizza_type_id, SUM(o.quantity) AS Total_Quantity
FROM
    pizzas p
        JOIN
    order_details o ON p.pizza_id = o.pizza_id
GROUP BY p.pizza_type_id;
```

pizza_type_id	Total_Quantity
hawaiian	2422
classic_dlx	2453
five_cheese	1409
ital_supr	1884
mexicana	1484
thai_ckn	2371
prsc_argla	1457
bbq_ckn	2432
the_greek	1420
spinach_supr	950
green_garden	997
ital_cpcllo	1438
spicy_ital	1924
spin_pesto	970
veggie_veg	1526

7.DETERMINE THE DISTRIBUTION OF ORDERS BY HOUR OF THE DAY.

```
SELECT
    HOUR(order_time) AS Hour, COUNT(order_id) AS Order_Count
FROM
    orders
GROUP BY HOUR(order_time);
```

Hour	Order_Count
11	1231
12	2520
13	2455
14	1472
15	1468
16	1920
17	2336
18	2399
19	2009
20	1642
21	1198
22	663
23	28
10	8
9	1

8.JOIN RELEVANT TABLES TO FIND THE CATEGORY-WISE DISTRIBUTION OF PIZZAS.

```
SELECT
    category, COUNT(name)
FROM
    pizza_types
GROUP BY category
ORDER BY COUNT(NAME) DESC;
```

category	COUNT(name)
Supreme	9
Veggie	9
Classic	8
Chicken	6

9.GROUP THE ORDERS BY DATE AND CALCULATE THE AVERAGE NUMBER OF PIZZAS ORDERED PER DAY.

```sql
SELECT
    ROUND(AVG(Quantity),0) AS Avg_Pizza_Ordered_Per_Day
FROM
    (SELECT
        o.order_date, SUM(od.quantity) AS Quantity
    FROM
        orders o
    JOIN order_details od ON o.order_id = od.order_id
    GROUP BY o.order_date) AS Order_Quantity;
```

Avg_Pizza_Ordered_Per_Day
138

10.DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE.

```
SELECT
    pt.name, ROUND(SUM(od.quantity * p.price), 2) AS Total_Sales
FROM
    pizzas p
        JOIN
    order_details od ON p.pizza_id = od.pizza_id
        JOIN
    pizza_types pt ON pt.pizza_type_id = p.pizza_type_id
GROUP BY pt.name
ORDER BY Total_Sales DESC
LIMIT 3;
```

name	Total_Sales
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5

11.CALCULATE THE PERCENTAGE CONTRIBUTION OF EACH PIZZA TYPE TO TOTAL REVENUE.

```
SELECT pt.category ,ROUND(((SUM(od.quantity*p.price)/ (SELECT
     SUM(od.quantity * p.price) AS Total_Sales
FROM
    pizzas p
        JOIN
    order_details od ON p.pizza_id = od.pizza_id))*100),2)
    AS Total_Revenue_in_Percentage
FROM
    pizzas p
        JOIN
    pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
    JOIN order_details od ON od.pizza_id=p.pizza_id
GROUP BY pt.category
ORDER BY Total_Revenue_in_Percentage DESC;
```

category	Total_Revenue_in_Percentage
Classic	26.91
Supreme	25.46
Chicken	23.96
Veggie	23.68

12.ANALYZE THE CUMULATIVE REVENUE GENERATED OVER TIME.

```
SELECT order_date ,
SUM(Revenue) OVER (ORDER BY order_date)
    AS Cumulative_Revenue FROM
    (SELECT o.order_date ,SUM(od.quantity*p.price)
    AS Revenue FROM
        order_details od
            JOIN
        pizzas p ON od.pizza_id=p.pizza_id
            JOIN
        orders o ON o.order_id=od.order_id
GROUP BY o.order_date) AS Sales;
```

order_date	Cumulative_Revenue
2015-01-01	2713.8500000000004
2015-01-02	5445.75
2015-01-03	8108.15
2015-01-04	9863.6
2015-01-05	11929.55
2015-01-06	14358.5
2015-01-07	16560.7
2015-01-08	19399.05
2015-01-09	21526.4
2015-01-10	23990.350000000002
2015-01-11	25862.65
2015-01-12	27781.7
2015-01-13	29831.300000000003
2015-01-14	32358.700000000004
2015-01-15	34343.50000000001
2015-01-16	36937.65000000001
2015-01-17	39001.75000000001
2015-01-18	40978.600000000006

THANK YOU